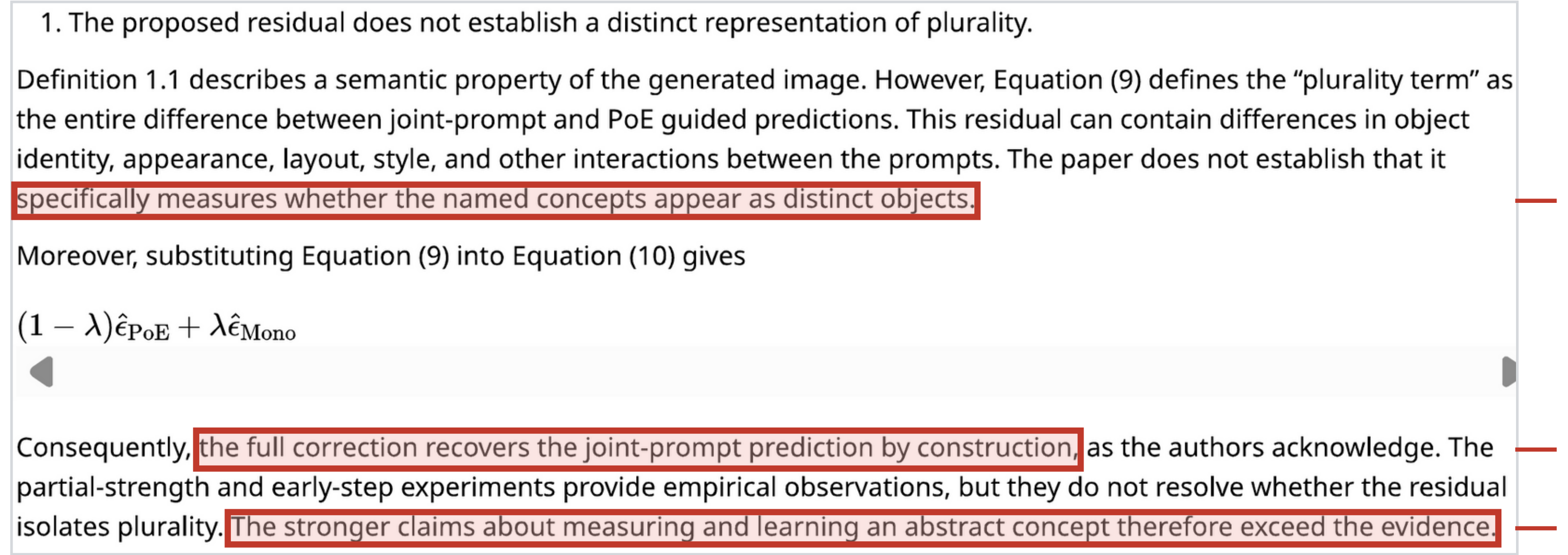
G1: What the residual measures

k4nd weakness 1 and question 1 · status: answered

THE REVIEWER'S POINT

k4nd, weakness 1



1. The proposed residual does not establish a distinct representation of plurality.

Definition 1.1 describes a semantic property of the generated image. However, Equation (9) defines the "plurality term" as the entire difference between joint-prompt and PoE guided predictions. This residual can contain differences in object identity, appearance, layout, style, and other interactions between the prompts. The paper does not establish that it specifically measures whether the named concepts appear as distinct objects.

Moreover, substituting Equation (9) into Equation (10) gives

$$(1-\lambda)\hat{\epsilon}_{\text{PoE}} + \lambda\hat{\epsilon}_{\text{Mono}}$$

Consequently, the full correction recovers the joint-prompt prediction by construction, as the authors acknowledge. The partial-strength and early-step experiments provide empirical observations, but they do not resolve whether the residual isolates plurality. The stronger claims about measuring and learning an abstract concept therefore exceed the evidence.

Reviews.pdf, page 4 (no text layer; passage located by OCR)

K-W1a

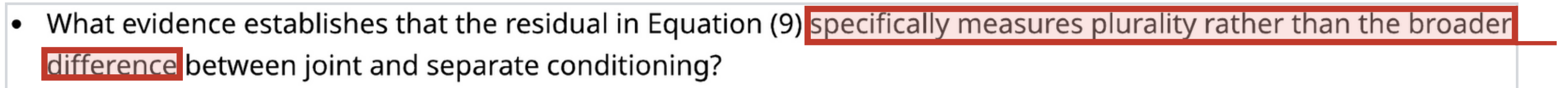
K-W1b

K-W1b

this claim

by construction

k4nd, question 1



- What evidence establishes that the residual in Equation (9) specifically measures plurality rather than the broader difference between joint and separate conditioning?

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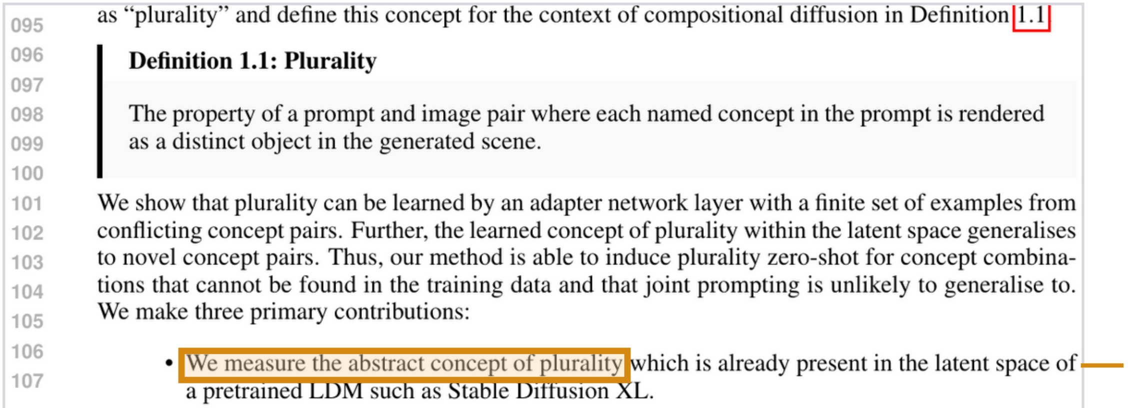
K-Q1

THE SUBMITTED WEAKNESS

Introduction, the contributions



as "plurality" and define this concept for the context of compositional diffusion in Definition 1.1

Definition 1.1: Plurality

The property of a prompt and image pair where each named concept in the prompt is rendered as a distinct object in the generated scene.

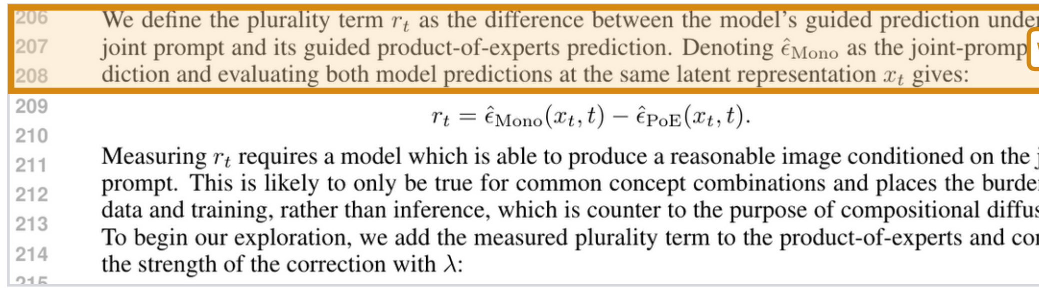
We show that plurality can be learned by an adapter network layer with a finite set of examples from conflicting concept pairs. Further, the learned concept of plurality within the latent space generalises to novel concept pairs. Thus, our method is able to induce plurality zero-shot for concept combinations that cannot be found in the training data and that joint prompting is unlikely to generalise to. We make three primary contributions:

- We measure the abstract concept of plurality which is already present in the latent space of a pretrained LDM such as Stable Diffusion XL.

56173_Restoring_plurality_in_c.pdf (submitted), page 2

measures plurality itself

Section 3, the plurality term



We define the plurality term r_t as the difference between the model's guided prediction under the joint prompt and its guided product-of-experts prediction. Denoting $\hat{\epsilon}_{\text{Mono}}$ as the joint-promp diction and evaluating both model predictions at the same latent representation x_t gives:

$$r_t = \hat{\epsilon}_{\text{Mono}}(x_t, t) - \hat{\epsilon}_{\text{PoE}}(x_t, t). \tag{9}$$

Measuring r_t requires a model which is able to produce a reasonable image conditioned on the joint prompt. This is likely to only be true for common concept combinations and places the burden on data and training, rather than inference, which is counter to the purpose of compositional diffusion. To begin our exploration, we add the measured plurality term to the product-of-experts and control the strength of the correction with λ:

56173_Restoring_plurality_in_c.pdf (submitted), page 4

whole difference, named plurality

now named for what it is

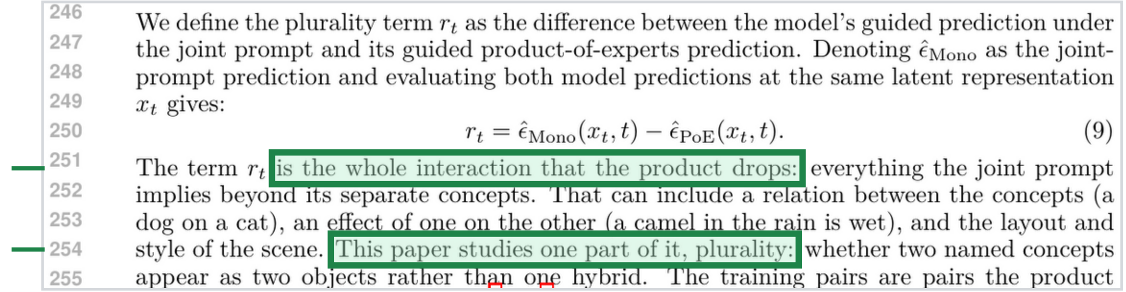
rewritten

THE REVISED ANSWER

Section 3, what the term is

the whole interaction

scoped to plurality



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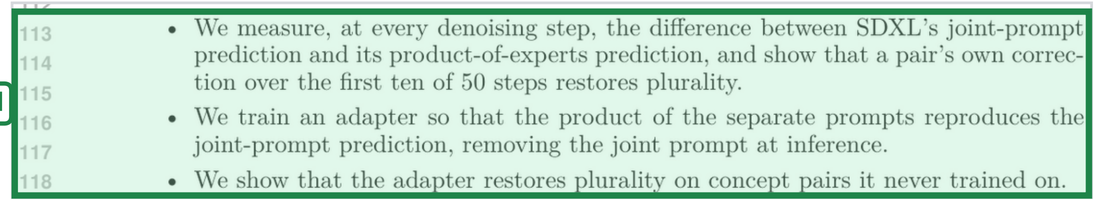
$$r_t = \hat{\epsilon}_{\text{Mono}}(x_t, t) - \hat{\epsilon}_{\text{PoE}}(x_t, t). \tag{9}$$

The term r_t is the whole interaction that the product drops: everything the joint prompt implies beyond its separate concepts. That can include a relation between the concepts (a dog on a cat), an effect of one on the other (a camel in the rain is wet), and the layout and style of the scene. This paper studies one part of it, plurality: whether two named concepts appear as two objects rather than one hybrid. The training pairs are pairs the product

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 5

Introduction, the contributions

claims only what is measured



- We measure, at every denoising step, the difference between SDXL's joint-prompt prediction and its product-of-experts prediction, and show that a pair's own correction over the first ten of 50 steps restores plurality.
- We train an adapter so that the product of the separate prompts reproduces the joint-prompt prediction, removing the joint prompt at inference.
- We show that the adapter restores plurality on concept pairs it never trained on.

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 3

The reply we plan to send (replies/k4nd-and-uhz6.md: k4nd W1 and Q1)

answered

W1: the residual does not establish a distinct representation of plurality

We agree that r_t in Equation 9 is the entire difference between the joint-prompt and product-of-experts predictions, and that adding it in full recovers the joint prompt by construction. We do not claim that r_t represents plurality alone. A product of experts drops the interaction the joint prompt carries, and that interaction can hold several kinds of meaning: a relation between the concepts ("a dog on a cat"), an effect of one on the other ("a camel in the rain" should be wet), and the layout and style of the scene.

Our study concerns one part of this interaction: plurality, whether two named concepts appear as two objects rather than one hybrid. The training pairs were chosen for it, as pairs the product fuses into a hybrid.

Within that scope, the evidence is that the correction is specific and early, not generic. A pair's own r_t restores the second object, while another pair's, another seed's, or a time-reordered one does not (Figure 2). The first ten of 50 steps are enough (Figures 3 and 4). At partial strength (λ of 0.6 to 0.65 in Figure 2) the second object appears while the layout and style still look like the product's render.

We agree this does not separate plurality from everything else r_t carries. Section 3 now states this scope explicitly, and the contributions are rewritten to claim only what the experiments show.

Q1: evidence that the residual measures plurality

Answered under W1 above.

Changes behind this page: Section 3 says r_t is the whole interaction the product drops and scopes the paper to plurality; the contributions claim only what the experiments show.

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